

ERIC LIU

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Education

University of Toronto

Sep 2022 – May 2027

Bachelor of Applied Science in Computer Engineering with PEY Co-op

Toronto, Ontario

- On track to graduate with Minors in Artificial Intelligence Engineering and Engineering Business
- Dean's List Scholar
- Relevant Courses: DSA, Software Communication and Design, Applied Fundamentals of Deep Learning, Networks

Technical Skills

Programming Languages: Python, C++, C, SQL, Java, JS, Verilog, MATLAB, Assembly, Angular, HTML/CSS

Frameworks & Tools: Stable Baselines3, PostgreSQL, PyTorch, Gym, Git, JSON, Docker, Linux, NumPy, AWS, GCP

Strong skills in advanced DSA: Graph Theory, Dynamic Programming, Monotonic Structures, Greedy Algorithms

Experience

Scotiabank - Quantitative Trading Developer

May 2025 – April 2026

Electronic Execution Services (Algorithmic Trading) - Global Banking and Markets

Toronto, Ontario

- **Tech Environment:** Bloomberg API, Transaction Cost Analysis, SQL, Python, Angular, JavaScript, Pandas.
- Completed the Outage App, a diagnostic tool integrated with Bloomberg to analyze failed algorithmic executions (e.g., market halts, volume limits, client caps), computed refund adjustments using trade-specific rates; optimized performance using vectorization and bitmasking for efficient calculations over extended trading outages.
- Investigated and resolved missing trade records in TCA pipelines; tracing data flow across upstream/downstream systems and refactoring reporting logic to improve data integrity and compliance accuracy.
- Maintained FX and Commission reconciliation pipelines; resolved trade discrepancies with middle office and traders.
- Developed and automated client-facing reporting tools, improving reporting efficiency and accuracy.

Big Tuna - Software Developer

Aug 2020 – May 2024

Seasonal Game Developer

Oakville, Ontario

- **Tech Environment:** JavaScript, JSON, SQL, Developer Portal, Game Design, Git, Software Testing.
- Part of a three-person team that created and maintained a highly engaging Discord bot with more than 23,000 players.
- Integrated SQL database updates corresponding to new content, and conducted comprehensive pre-deployment testing.

Research

Market Making Via Reinforcement Learning Research

May 2023 – Aug 2024

Research Assistant Under Professor Wei Xu

Toronto, Ontario

- **Tech Environment:** Python, Stable Baselines (DDPG and DQN algorithms), Linux, Pandas, Gym, PyTorch.
- Researched and developed a profitable market-making agent for cryptocurrency exchange using Deep Deterministic Policy Gradient (DDPG) and Deep Q-Network (DQN), understanding its underlying mathematical and financial constructs.
- Crafted a sophisticated machine learning environment that mimics real-world market conditions and scenarios, integrating over 100 million tradebook data entries, paired with a dynamic reward framework that reacts and gives appropriate response to the model's decisions, facilitating continuous learning and strategic optimization.

Extracurricular

University of Toronto Aerospace Design Team

Jul 2023 – Present

Liquid Propulsion and Aerodynamics Division in Rocketry

Toronto, Ontario

- **Tech Environment:** Python, Linux, NumPy, Rocketcea, Git, Project Management .
- Completing code rework that simulates flight data and predicts results, transitioning an outdated 2D point system with two degrees of freedom (2DoF) to a more accurate 3D model with six degrees of freedom (6DoF).

Ontario Engineering Competition (OEC)

January 2024

Represented the University of Toronto for programming competition

Kingston, Ontario

University of Toronto Engineering Kompetition (UTEK)

November 2023

First place in programming competition (Team of 4)

Toronto, Ontario

Projects

Fungi Identification Project using Deep Learning | Python, Git, PyTorch, NumPy, Pandas

Jul 2024

- Developed a comprehensive Convolutional Neural Network (CNN) model that identifies various species of fungi, trained on over 10,000 images, achieving model accuracy of 81% compared to baseline accuracy of 65%.

GIS Mapping System | C++, GTK, Git, Multithreading, Heuristic Algorithms

May 2024

- Implemented map functionalities, graphic rendering, and multithreading optimizations, and utilized time precomputation methods to enhance performance, reducing time complexity.
- Solved a variant of the Traveling Salesman Problem using heuristic techniques for route optimization.